Cloud computing

Course notes

Daniele Besozzi

A.A. 26/27

Contents

1 Introduction and definition 2
1.1 Essential characteristics 2
1.2 Why cloud computing? 3
1.3 Data centers and cloud computing 3
1.4 Virtual infrastructure manager (VIM) 4
2 Service models (SPI) 4
2.1 Infrastructure as a Service (IaaS) 4
2.1.1 Resource Management Interface 5
2.1.2 System Monitoring Interface 5
2.1.3 Properties of IaaS 5
2.2 Platform as a Service (PaaS) 6
2.2.1 Properties of PaaS 7
2.3 Software as a Service (SaaS) 8
2.3.1 Properties of SaaS 9
2.4 The SPI model 9
3 Deployment models 10
3.1 Public Cloud 10
3.2 Private Cloud 10
3.3 Community Cloud 11
3.4 Hybrid Cloud 11

1 Introduction and definition

We can define cloud computing in many ways, but the most important thing is to understand the basic definition. A Cloud is a type of distributed system consisting of:

- **interconnected** and **virtualized** computers
- **provisioned on demand** and
- **accessible** through a network.

Alternatively, we can say that cloud computing is:

- a **user experience**, in a sense that it is a way to provide services to users over the network, and
- an **infrastructure management methodology**, because it is a way to manage large numbers of highly virtualized resources so that they can be automatically aggregated to deliver services.

We can give a more formal definition by the NIST (National Institute of Standards and Technology):

> **Definition 1.1** *(Cloud computing).* Cloud computing is a **model** for enabling **convenient**, **on-demand network access** to a **shared** pool of **configurable computing resources** (e.g., networks, servers, storage, applications, and services) that can be **rapidly provisioned and released with minimal management** effort or service provider interaction.

This cloud model promotes availability and is composed of:

- five essential characteristics,
- three service models, and
- four deployment models.

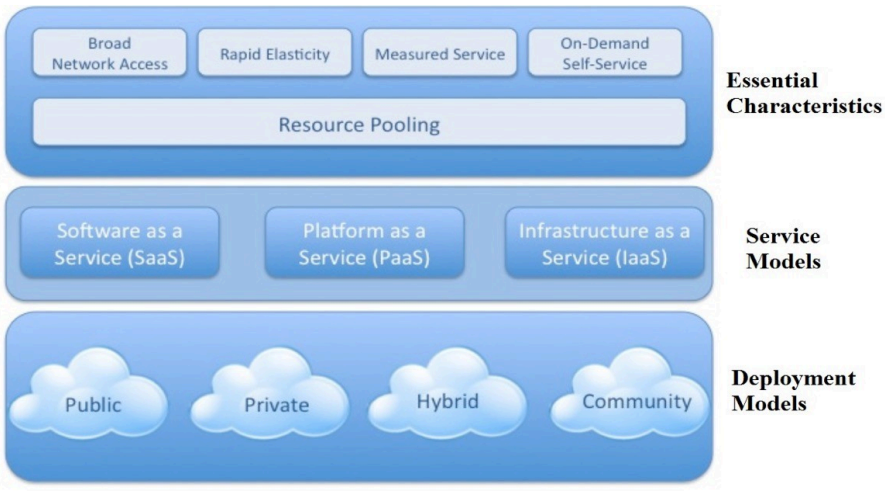


Figure 1: Cloud computing characteristics and models

This gives us a technological, economical, and societal model.

1.1 Essential characteristics

The essential characteristics of cloud computing are:

- **On-demand self-service**:
  - A consumer can unilaterally provision computing capabilities such as server time and network storage as needed automatically, without (or with little) requiring human interaction with a cloud provider.
- **Broad network access**:

    - Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and PDAs) as well as other traditional or cloud- based software services.
- **Resource pooling**:
    - The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
- **Rapid elasticity**:
    - Capabilities can be rapidly and elastically provisioned to quickly scale up; and rapidly released to quickly scale down, automatically.
    - To consumers, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.
- **Measurable service**:
    - Cloud systems automatically control and optimize resource usage by leveraging a metering capability at some level of abstraction appropriate to the type of service.
    - Resource usage can be monitored, controlled, and reported - providing transparency for both the provider and consumer of the service.

1.2 Why cloud computing?

The cloud shape is often used in network diagrams to represent a "black box" network. The access **interfaces are known, but not the internal structure**. A cloud is opaque and considered very large and distant. As in the sky there are **different types of clouds** so there are **different types of services** and **features** offered in the cloud. The internal implementation of these services is **opaque**.

Clouds may overlap, that is, different models of clouds can be implemented by the same vendor. Clouds can **dynamically change shape**. Applications and services in cloud can dynamically **adapt**.

1.3 Data centers and cloud computing

Data centers are buildings that house hundreds of thousands of servers that simultaneously support many services and applications. Data centers are built on a large scale to reduce maintenance costs, but not very large (also to limit blast radius).

Data centers exploit **virtualization** massively to run applications.

A data center requires:

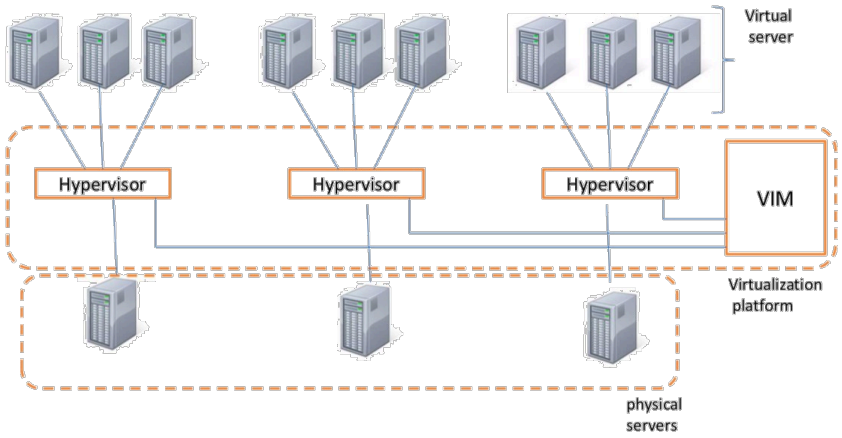
- **Power supply** (generator system to overcome possible power failures).
- Continuous **cooling** system.
- **Security system** (fire alarms, adequate ventilation to eliminate smoke, authorized access, video cameras, security code access, biometric access, card reader access, locked cabinets…).
- **High-speed network**, within the data center, between data centers and to the Internet Often optical fiber, dedicated and proprietary lines, including intercontinental lines.

> **Definition 1.2** *(Availability zone).* Every Availability Zone is an independent data center with its own power grid and network so that it is protected from possible failures in other data centers

Definition 1.3 *(Region).* A Region is a set of Availability Zones that are geographically close to each other and connected by low-latency links. A Region is isolated from other Regions, so that a failure in one Region does not affect other Regions

1.4 Virtual infrastructure manager (VIM)

A key challenge cloud providers face in building an infrastructure is managing physical and virtual resources, i.e., servers, storage and networks, holistically. **Resource orchestration** must be performed in a way that quickly and dynamically delivers resources to users and applications. The software tool responsible for the orchestration is called a **Virtual Infrastructure Manager (VIM)**. This type of software resembles a traditional operating system ("cloud operating system") - but instead of dealing with a single computer, it aggregates resources from multiple computers, presenting a uniform view to the user and applications.



2 Service models (SPI)

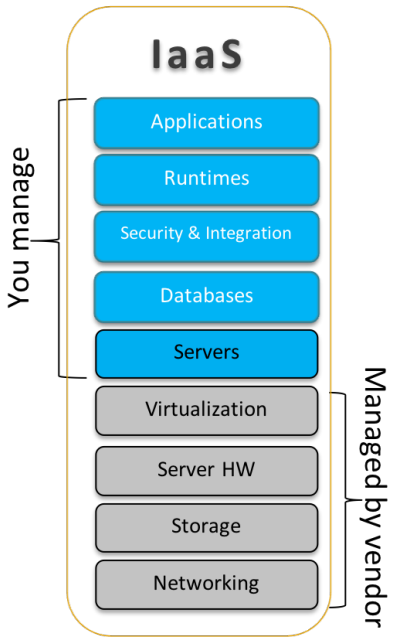
A service model is, in this scenario, a high-level characterization (model) of the cloud that **focuses primarily on the services offered and the type of users**. The set of service models defines a possible **taxonomy**.

2.1 Infrastructure as a Service (IaaS)

The capability provided to the consumer (user or application) is to provision:

- virtual machines,
- storage,
- networks, and
- other fundamental computing resources

The consumer can deploy and run arbitrary software on the virtualized resources, which can include OS and applications. The consumer **does not manage or control the underlying cloud infrastructure** but has **control over operating systems, storage, deployed applications**, and possibly limited control of selected networking components.



This kind of technique is enabled by **virtualization**.

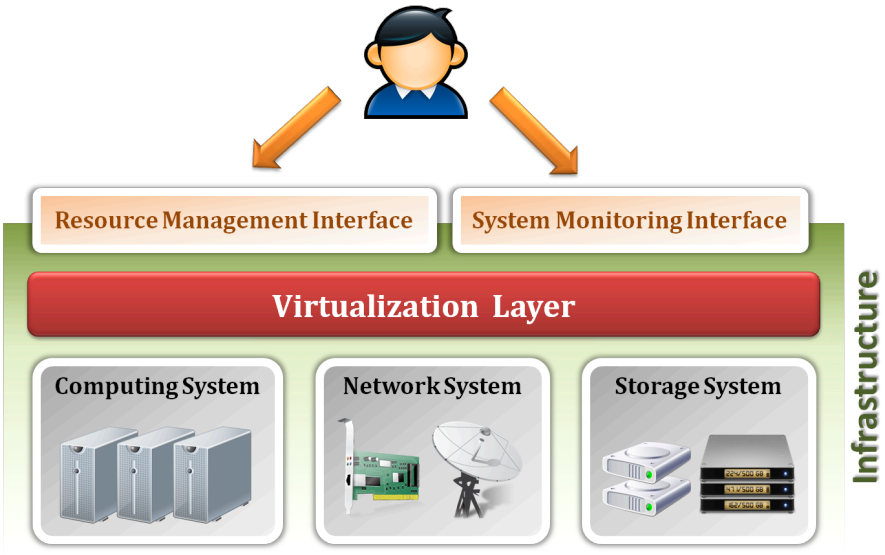


Figure 4: IAAS architecture

2.1.1 Resource Management Interface

RMI Enables the customer to obtain and manage several types of virtualized resource:

- **Virtual Machine**: IaaS provides the basic virtual machine operations, such as creation,, suspension, resumption and termination.
- **Virtual Storage**: IaaS provides the basic virtual storage operations, such as space allocation, space release, data writing and data reading.
- **Virtual Network**: IaaS provides the basic virtual network operations, such as IP address allocation, domain name register, connection establishment and bandwidth provision.

2.1.2 System Monitoring Interface

There are several metrics to be monitored:

- **Virtual Machine**: As an IaaS provider, we should be able to monitor some system states of each virtual machine, such as CPU loading,, memory utilization, IO loading and internal network loading
- **Virtual Storage**: As an IaaS provider, we should be able to monitor some storage states of each virtual storage, such as virtual space utilization, data duplication and storage device access bandwidth, ...etc.
- **Virtual Network**: As an IaaS provider, we should be able to monitor some network states of each virtual network, such as virtual network bandwidth, network connectivity and network load balancing

2.1.3 Properties of IaaS

- Manageability and Interoperability
  - System control, billing, API
- Availability and Reliability
  - Multiple copies of VMs, storage redundancy
- Scalability and Elasticity
  - On demand resource (ex. VMs) provisioning

> This technique is enabled by **virtualization**



Figure 5: Properties of IaaS

2.2 Platform as a Service (PaaS)

The capability provided to the consumer is to **develop and deploy** onto the cloud applications created using **programming languages and tools supported by the provider**. The **consumer** does not manage or control the underlying cloud infrastructure including network, servers, operating systems, or storage, but has **control over the deployed applications and possibly application hosting environment configurations**.

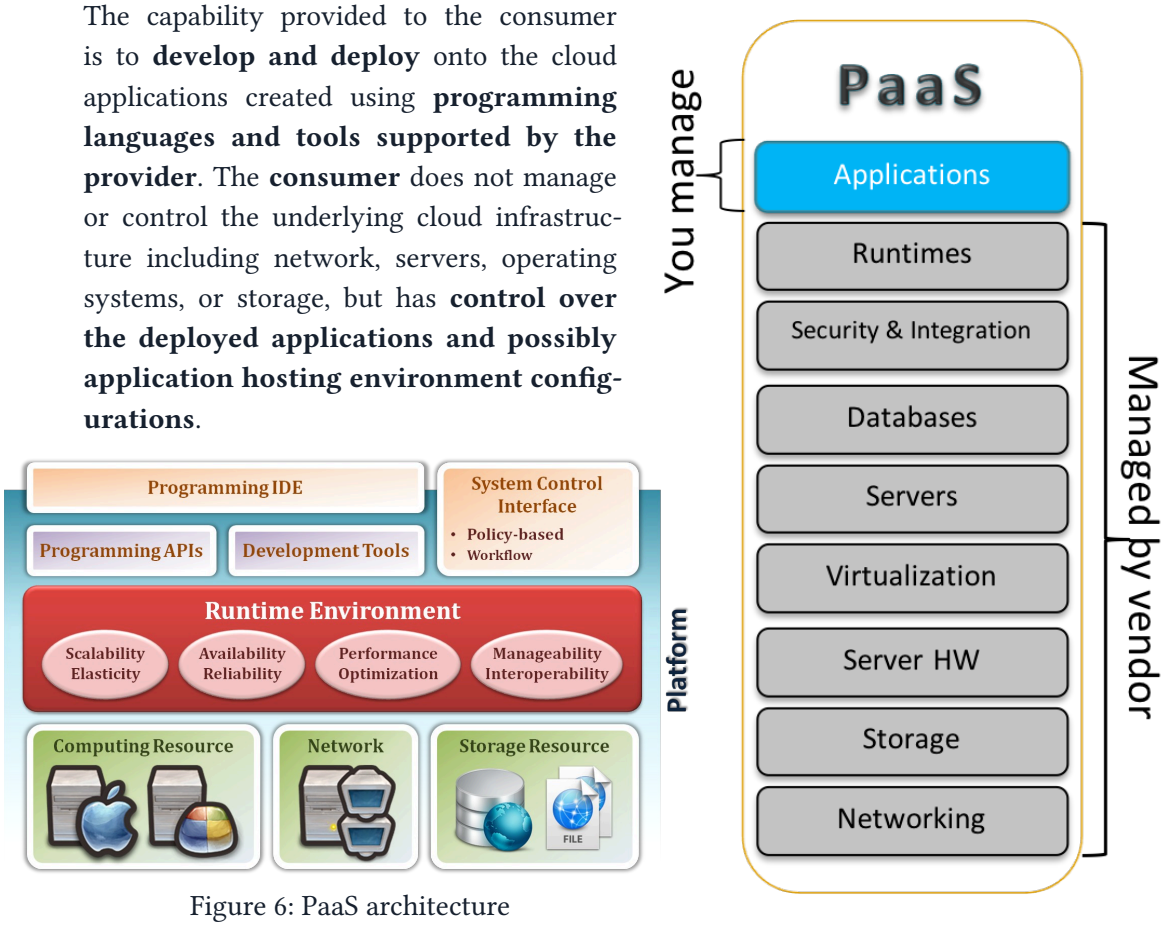


Figure 6: PaaS architecture

> This technique is enabled by **runtime environment**, that is acollection of software services available. Usually implemented by a collection of program libraries and an application execution engine

PAAS offers different services:

- Users make use of **programming IDE** to **develop their service** over PaaS.
  - This IDE should integrate the full functionalities supported from the underling runtime environment. This IDE should also provide development tools, such as profiler, debugger and testing environment.
  - The programming APIs supported from runtime environment may be heterogeneous across different cloud providers, but there are still some common operating functions.
- **System Control Interface**
  - **Policy-Based Control**
    - Typically described as a principle or rule to guide decisions and achieve rational outcome(s)
    - Make the decision according to some requirements
    - If <condition> then <action>
  - **Workflow Control**
    - Describes the flow of installation and configuration of resources
    - Delivers speedy and efficient construction and management of cloud resources

2.2.1 Properties of PaaS

Common properties in Runtime Environment :

- Manageability and Interoperability
- Performance and Optimization
- Availability and Reliability
- Scalability and Elasticity



Figure 8: Properties of PaaS

2.3 Software as a Service (SaaS)

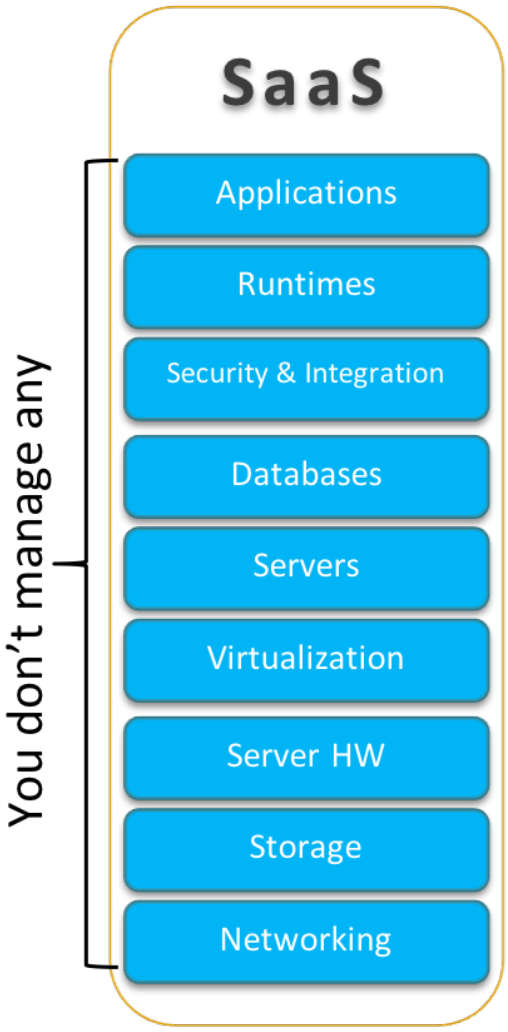
The capability provided to the consumer is to use the **provider's applications running on a cloud infrastructure**. The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web- based email).

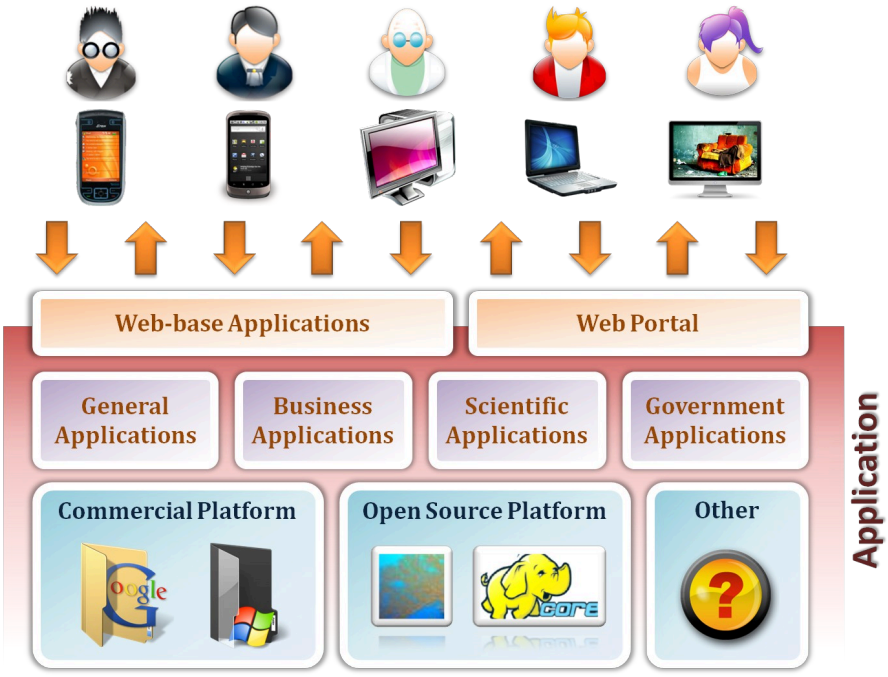
The **consumer does not manage or control the underlying cloud infrastructure** including network, servers, operating systems, storage, or individual application capabilities, except for limited user-specific application configuration settings.

> This technique is enabled by **web services**

Web 2.0 is the trend of using the full potential of the web:

- Viewing the Internet as a computing platform
- Running interactive applications through a web browser
- Leveraging interconnectivity and mobility of devices
- Enhanced effectiveness with greater human participation





Services can be provided in many forms:

- **web-base applications**. Conventional applications should translate their access interface onto web-based platform. Applications can have different domains:
  - General Applications : Applications which are designed for general propose, such as office suit, multimedia and

  instant message.
  - Business Applications : Application which are designed for business propose, such as ERP, CRM and market

  trading system.
  - Scientific Applications : Application which are designed for scientific propose, such as aerospace simulation and

  biochemistry simulation.
  - Government Applications : Applications which are designed for government propose, such as national medical

  system and public transportation system service.
- **Web portal**: Apart from the standard search engine feature, web portals offer other services such as e-mail, news, stock prices, information, databases and entertainment. Portals provide a way for enterprises to provide a consistent look and feel with access control and procedures for multiple applications and databases, which otherwise would have been different entities altogether.

2.3.1 Properties of SaaS

The properties offered by SaaS are the ones offered by the internet, that is **accessibility** and **portability**.

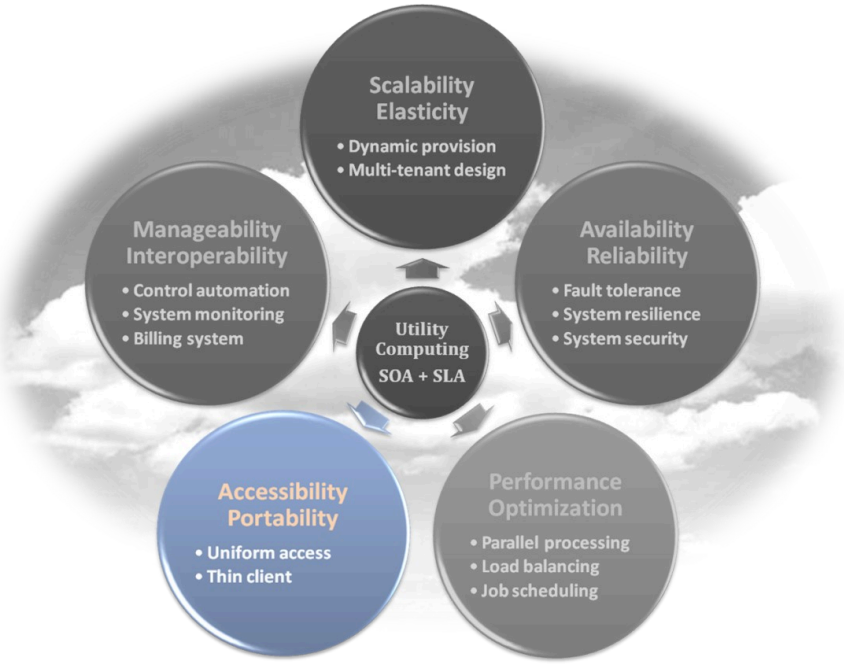


Figure 11: Properties of SaaS

2.4 The SPI model

The SPI model (SaaS, PaaS, IaaS) is the most common classification of the cloud services. But, be aware that the **real cloud is blurry**. The provider can offer different services at different levels, and it is not always easy to classify them.

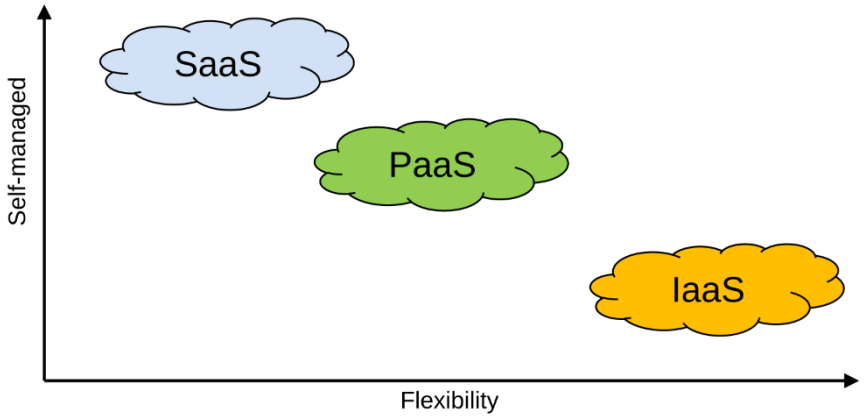


Figure 12: The SPI model

3 Deployment models

A cloud deployment model is defined according to **where the infrastructure resides** and **who has control over that infrastructure**. Deciding which deployment model you will go with is one of the most important cloud deployment decisions you will make.

There are four primary cloud models:

- Public Cloud
- Private Cloud
- Community Cloud
- Hybrid Cloud

Differences lie primarily in the **access**, **size** and **ownership** of published cloud services

3.1 Public Cloud

The cloud infrastructure is made **available to the general public** and is owned by an **organization selling cloud services**. Also known as external cloud or multi-tenant cloud, this model essentially represents a cloud environment that is openly accessible. Basic characteristics:

- **Homogeneous** infrastructure
- Common policies
- Shared resources and multi-tenancy
- Leased or rented infrastructure
- Economies of scale

3.2 Private Cloud

The cloud infrastructure is **operated solely for one organization**. It may be managed by the same organization or a third party and may exist **on premise or off premise**.

Also referred to as internal cloud or on-premise cloud, a private cloud intentionally limits access to its resources to consumers that belong to the organization that owns the cloud. Basic characteristics :

- **Heterogeneous** infrastructure
- Customized and tailored policies
- Dedicated resources
- In-house infrastructure
- End-to-end control

3.3 Community Cloud

The cloud infrastructure is **shared by several organizations** It supports a specific community that has shared concerns (e.g., mission, security requirements, policy, and compliance considerations). Access to a community cloud environment is **typically restricted to the members of the community**.

3.4 Hybrid Cloud

The cloud infrastructure is a **composition** of two or more clouds (private, community, or public) that remain unique entities but are **bound together by standardized or proprietary technology** that enables data and application portability (e.g., cloud bursting for load-balancing between clouds). Typical example: Private-Public for scalability.